MIL-F-60543A(MU)
14 October 1966
SUPERSEDING
MIL-F-60543(MU)
14 June 1966

MILITARY SPECIFICATION

FUZE, MT, M565E1, ASSEMBLY

1. SCOPE

1.1 This specification covers Fuze, MT, M565E1 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-314 - Waterproofness Test for Use in Development of Fuzes
MIL-STD-350 - Jolt Test for Use in Production of Fuzes
MIL-STD-351 - Jumble Test for Use in Production of Fuzes
MIL-STD-352 - Forty-Foot Drop Test for Use in Production of Fuzes
MIL-STD-353 - Transportation-Vibration Test for Use in Production of Fuzes
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

C8861213 - Box, Wirebound Packing, Ammunition for Artillery Type and Rocket Fuzes (within M2A1 Box)
F8864492 - Box, Ammunition, Metal for Artillery Type and Rocket Fuzes
F10542897 - Fuze, MT, M565E1 Assembly
IEL-10542897 - Inspection Equipment List for Fuze M565E1 Assembly

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications except as provided in 6.1.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.3 Dynamic regulations.- The movement subassembly shall be regulated to the frequency specified on the applicable drawing. The timing error between 2 to 100 seconds shall not exceed 0.07 percent while spinning at the specified revolutions per minute (rpm).

3.4 Environmental

3.4.1 Jolt.- The fuze shall comply with MIL-STD-350.

3.4.2 Jumble.- The fuze shall comply with MIL-STD-351.

3.4.3 Forty-foot drop.- The fuze shall comply with MIL-STD-352.

3.4.4 Waterproofness.- The fuze shall comply with MIL-STD-314.

3.4.5 Transportation-vibration.- The fuze shall comply with MIL-STD-353.

3.5 Ballistics (see 6.3.4)

3.5.1 The fuze shall not function prematurely (see 6.3.1)

3.5.2 Target impact nonfunction distance.- The fuze with point detonating assembly (see IEL) shall not detonate the round upon impact element function when fired in the 105MM Howitzer, charge 7 (1550 feet per second, fps) with the fuze set safe "S" on a target 220 plus 5 feet from the muzzle of the weapon.

3.5.3 Time-to-burst (ambient temperature).- The fuze shall detonate the round during flight within the time limits specified in Table I, when fired as follows:

Weapon	Charge	Setting
105MM Howitzer	7 (1550 fps)	60 seconds
155MM Howitzer	1 (680 fps)	3 seconds

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3.5.4 Excess pressure.- The fuze shall detonate the round during flight when fired in the 105MM Howitzer at 112% of the normal charge 7 pressure with the fuze set at 60 seconds.

3.5.5 Low temperature.- The fuze temperature conditioned at minus 65 to minus 70 degrees Fahrenheit (F) shall detonate the round during flight when fired as specified in 3.5.3.

3.5.6 High temperature.- The fuze temperature conditioned at plus 145 to plus 150 degrees F shall detonate the round during flight when fired as specified in 3.5.3.

TABLE I

Limits for Sample Mean Time of Flight (Sec)

Weapon	Charge	Fuze Setting	Limits for Individuals (Seconds)		Sample Size 20	19	18	17	16	15
155MM How.	1	3 sec	UL	3.48	3.253	3.255	3.257	3.259	3.261	3.263
			LL	2.88	3.105	3.103	3.101	3.099	3.097	3.095
105MM How.	7	60 sec	UL	60.50	59.906	59.911	59.916	59.921	59.926	59.931
			LL	58.90	59.498	59.493	59.488	59.483	59.478	59.473

Maximum Limits for Sample Standard Deviation (sec)

Weapon	Charge	Fuze Setting	Sample Size 20	19	18	17	16	15
155MM How.	1	3 sec	.133	.134	.135	.137	.138	.139
105MM How.	7	60 sec	.369	.372	.375	.378	.382	.386

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

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3.6.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.6.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.6.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

320 component parts as delineated in 4.2.2
450 subassemblies as delineated in 4.2.3
131 fuze assemblies as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection by the contractor and provisional acceptance at source by the Government, 105 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Three

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4.2.1.1 (Cont'd)

hundred and twenty (320) component parts, 450 subassemblies and 26 fuze assemblies shall be inspected as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components

4.2.2.1 Twenty (20) of each component shall be tested as in 4.3.3.1 (Hardness), 4.3.3.2 (Load) and 4.3.3.4 (Bend).

4.2.3 Subassemblies

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.5.1 (static load).

4.2.3.2 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.5.2 (torque).

4.2.3.3 Twenty-five (25) of each assembly shall be tested as in 4.3.3.5 (pull test).

4.2.4 Acceptance criteria.- Failure of any component or subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the initial production sample.

4.2.5 Fuze assemblies. The twenty-six (26) fuze assemblies shall be subjected to the following MIL-STD tests, with the fuzes set on safe "S".

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.4.1	4.4.17.1	4.2.5.1.1
Jumble	4	3.4.2	4.4.17.2	4.2.5.1.1
Forty-Foot Drop	5	3.4.3	4.4.17.3	4.2.5.1.2
Waterproofness	8	3.4.4	4.4.17.4	4.2.5.1.3
Transportation-Vibration	5	3.4.5	4.4.17.5	4.2.5.1.4

4.2.5.1 Acceptance criteria.- Occurrence of any of the following in any fuze shall be cause for rejection of the initial production sample.

4.2.5.1.1 Jolt and jumble

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4.2.5.1.1 (Cont'd)

a. Any of the explosive elements burned or detonated

b. Rotor of safety adapter armed or partially armed (see 6.2.1)

c. Firing pin in fired position

d. Movement has run off safe position, i.e., drop-off edge of firing arm is not within scribed lines on safety disc.

e. Timing disc retainer loose

f. Broken, deformed or displaced parts which render the fuze unsafe to handle or dangerous to use.

4.2.5.1.2 Forty (40) foot drop

a. Safety adapter armed or partially armed

b. Rotor detonator burned or detonated

4.2.5.1.3 Waterproofness

a. Evidence of water entry

NOTE: Evidence of water entry through the safety adapter flash hole does not constitute reason for rejection.

4.2.5.1.4 Transportation-vibration

a. Conditions listed under 4.2.5.1.1

b. Failure of any movement to meet the nonfunction or function requirements or safety adapter to meet the nonarming or arming requirements of the applicable drawings.

4.2.6 Ballistics. One hundred and five (105) of the initial production sample, shall be subjected to the following ballistic test phases.

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4.2.6 (Cont'd)

Phases	Quantity	Requirement	Test Method
Impact Nonfunction Distance	20	3.5.1 & 2	4.4.18.1
Time-to-Burst (ambient temperature)			
155MM Howitzer charge 1	20	3.5.1 & 3	4.4.18.2
105MM Howitzer charge 7	20	3.5.1 & 3	4.4.18.2
Excess Pressure	25	3.5.1 & 4	4.4.18.3
Low Temperature (-65 to -70°F)			
155MM Howitzer charge 1	5	3.5.1 & 5	4.4.18.4
105MM Howitzer charge 7	5	3.5.1 & 5	4.4.18.4
High Temperature (+145° to +150°F)			
155MM Howitzer charge 1	5	3.5.1 & 6	4.4.18.4
105MM Howitzer charge 7	5	3.5.1 & 6	4.4.18.4

4.2.6.1 Acceptance criteria.- The occurrence of any of the following shall be cause for rejection of the initial production sample:

a. A premature burst in any phase; (see 6.3.1)

b. Function of one or more rounds upon target impact in the Target Impact Nonfunction Distance Phase.

c. Any of the following in the ambient Time-to-Burst Phase at each time setting.

(1) More than one time failure (6.3.2)

(2) More than one off times (6.3.3)

(3) Failure of the sample statistics to comply with the limits specified in Table I.

d. More than two time failures (6.3.2) in Excess Pressure Phase

e. More than two time failures (6.3.2) in either of the Environmental Phases (low temperature, high temperature).

4.3 Inspection provisions

4.3.1 Submission of product.- The product shall be presented in accordance with Standard Mil-STD-105.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All non-conforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing examination for visual defects.

4.3.2.1 Plate No. 1 first lamina Drawing F10523181, covering a detail of Drawing F10542897.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred, damaged or defective	Visual
105.	Chamfer of holes M and 20 missing or incorrect	Visual

4.3.2.2 Plate No. 2 second lamina Drawing F8595968, covering a detail of Drawing F10542897.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred, damaged or defective	Visual
105.	Chamfer on holes M and 20 missing or incorrect	Visual

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4.3.2.3 Plate No. 2 first lamina Drawing F8595967, covering a detail of Drawing F10542897.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	AQL = 1.0%
101. Flatness incorrect	Gage
102. Metal burred, damaged or defective	Visual

4.3.2.4 Slide, pallet lock Drawing C10521169, covering a detail of Drawing F10542897.

CRITICAL	
1. Radius on extension above max	Gage
MAJOR	
None defined	

4.3.2.5 Gear Escapement Drawing D10542765, covering a detail of Drawing F10542897.

CRITICAL	
None defined	
MAJOR	AQL = 1.0%
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

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4.3.2.6 Escapement gear and pinion assembly Drawing C10542766, covering a detail of Drawing F10542897.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Eccentricity of gear with datum	Gage
102.	Perpendicularity of gear, incorrect	Gage
103.	Burr	Visual

4.3.2.7 Escapement lever assembly Drawing F10542852, covering a detail of Drawing F10542897.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Eccentricity of external radius with datum A including parallelism with datum A	Gage
102.	Metal burred, damaged or defective	Visual
103.	Surface finish improper	Visual

4.3.2.8 Arbor Escapement Drawing D10535526, covering a detail of Drawing F10542897.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Perpendicularity of lever seat with datum C, incorrect	Gage
102.	Diameter of pivots, incorrect	Gage
103.	Metal burred, damaged or defective	Visual

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4.3.2.9 Detonator screw assembly Drawing C10520294, covering a detail of Drawing F10542897.

Defect	Method of Inspection
CRITICAL	
1. Detonator extends above flush of detonator screw	Visual
MAJOR AQL = 1.0%	
101. Overall length exceeds maximum	Gage
102. Insecure, improper or incomplete assembly	Manual & Visual
103. Green end of detonator not at bottom (thread end)	Visual
104. Crimping inadequate	Visual

4.3.2.10 Firing arm assembly, Drawing C10521115, covering a detail of Drawing F10542897.

CRITICAL	
1. Fracture of upright bend of firing arm	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 1.0%	
101. Insecure weight	Manual & Visual
102. Improper or incomplete assembly	Visual
103. Metal burred, damaged or defective	Visual

4.3.2.11 Cap, lower Drawing F10520255, covering a detail of Drawing F10542897.

CRITICAL	
None defined	
MAJOR AQL - 1.0%	
101. Diameter incorrect at forward datum length	Gage
102. Diameter incorrect at rear datum length	Gage
103. Eccentricity of forward diameter with datum A	Gage
104. Eccentricity of rear diameter with datum A	Gage

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4.3.2.12 Head, Drawing D10520795, covering a detail of Drawing F10542897.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter incorrect at forward datum length	Gage
102. Diameter incorrect at rear datum length	Gage
103. Eccentricity of forward diameter with datum A	Gage
104. Eccentricity of rear diameter with datum A	Gage

4.3.2.13. Plug, closing Drawing F10520253, covering a detail of Drawing F10542897.

CRITICAL	
1. Small O.D. below min	Gage
MAJOR AQL = 1.0%	
101. Location of locating pin hole incorrect	Gage

4.3.2.14 Body Drawing F8595997, covering a detail of Drawing F10542897.

CRITICAL	
1. Major diameter of large external thread, improper	Gage
MAJOR AQL = 1.0%	
101. Pitch diameter of large external thread, improper	Gage
102. Diameter of cavity exceeds max.	Gage
103. Threads damaged or not full	Visual
104. Metal burred, damaged or defective	Visual

4.3.2.15 Body Drawing F10542901, covering a detail of Drawing F10542897.

CRITICAL	
1. Minor diameter of adapter cavity thread exceeds max.	Gage

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Pitch diameter of adapter cavity thread, incorrect	Gage
102. External diameter incorrect at forward datum	Gage
103. External diameter incorrect at rear datum	Gage
104. Location of closing plug locating hole incorrect	Gage
105. Threads damaged or not full	Visual

4.3.2.16 Firing pin, Drawing C10521154, covering a detail of Drawing F10542897.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Diameter of striker point, improper	Gage
102. Surface finish on angle, improper	Visual

4.3.2.17 Timing disc assembly, Drawing D10520268, covering a detail of Drawing F10542897.

CRITICAL

1. Safety disc tabs not engaged with timing disc bushing slots	Visual

MAJOR AQL =0.40%	
101. Adjusting segment radius extends beyond periphery of timing disc	Visual
102. Safety disc radius below periphery of timing disc	Visual
103. Improper or incomplete assembly	Visual

4.3.2.18 Movement subassembly (after dynamic regulation), Drawing F10523481, covering a detail of Drawing F10542897.

CRITICAL

1. Firing pin fires on "not fire" position of firing arm	Gage
2. Extension of firing arm upright above top plate less than minimum (with firing arm shaft shoulder bottomed on plate No. 2)	Gage
3. Escapement gear not locked in properly by lever pallet	Visual

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4.3.2.18 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Extension of firing arm upright above top plate exceeds maximum (with firing arm shaft shouldered against top plate)	Gage
102. Firing pin does not fire on "must fire" position of firing arm	Gage
103. Firing pin protrusion less than min., in fired position	Gage
104. Firing arm shaft spring protrudes	Visual
105. Insecure, improper, or incomplete assembly	Manual & Visual
106. Staking of body assembly screws inadequate (check prior to regulation)	Visual
107. Arbor center hole missing	Visual

4.3.2.19 Movement assembly, Drawing F10523480, covering a detail of Drawing F10542897.

CRITICAL	
1. Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual & Visual
2. Timing disc retainer insecure, damaged or missing	Manual & Visual
3. Distance from top of top plate to top of retainer exceeds maximum (after staking of retainer)	Gage
MAJOR AQL = 0.40%	
101. Total height exceeds max.	Gage
102. Insecure, improper or incomplete assembly	Visual
103. Escapement lever not locked on proper side of pallet lock slides	Visual
104. Escapement spring broken, deformed or unseated	Visual
105. Foreign matter	Visual
106. Pins or protrusions above flush of bottom surface	Visual
107. Firing pin striker point damaged	Visual

4.3.2.20 Body, movement and lower cap assembly, Drawing F10542899 (prior to insertion of welsh plug), covering a detail of Drawing F10542897.

CRITICAL	
1. Drop off edge of firing arm outside of lines on safety disc (to be checked after lockring tensioning and zero setting operations)	Visual

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4.3.2.20 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Dimension from locknut to top of cap less than min	Gage
102. Eccentricity of arbor to datum "A"	Gage
*103. Setting pin extension below timing disc lug less than minimum	Gage
104. Insecure, improper or incomplete assembly	Manual & Visual
*105. Setting pin not engaged in timing disc lug slot	Visual
106. Body assembly screw(s) inadequately staked	Visual

*To be checked prior to assembly of the lower cap.

4.3.2.21 Fuze subassembly, less safety adapter, prior to assembly of closing plug assembly, Drawing F10542898, covering a detail of Drawing F10542897.

CRITICAL	
1. Escapement lever pallet(s) missing, damaged or deformed	Visual
MAJOR AQL = 0.40%	
101. Detonator screw above flush of body surface	Visual
102. Detonator screw assembly inserted improperly	Visual
103. Staking of detonator screw assembly inadequate	Visual

4.3.2.22 Fuze subassembly, less safety adapter, Drawing F10542898, covering a detail of Drawing F10542897 with closing plug assembly inserted.

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Distance from flange to top exceeds max.	Gage
102. Insecure, improper or incomplete assembly	Manual & Visual
103. Relay improperly inserted in closing plug (closed end not visible)	Visual
104. Relay extends beyond flush	Visual
105. Spinning or crimping of relay inadequate	Visual
106. Staking of assembly screws inadequate	Visual
107. Metal burred, damaged or defective	Visual

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4.3.2.23 Rotor assembly Drawing F8595998, covering a detail of Drawing F10542897.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Insecure assembly	Manual
102. Detonator installed incorrectly (with yellow end at crimp)	Visual
103. Detonator extends above flush of rotor surface	Visual
104. Rotor gear pin extends above flush of rotor surface	Visual
105. Staking or crimping of detonator assembly inadequate	Visual
106. Metal burred, damaged or defective	Visual

4.3.2.24 Safety adapter assembly, Drawing F8596049, covering a detail of Drawing F10542897.

CRITICAL	
1. Interference condition which precludes return of either spring-loaded detent from the fully open position to the position of engagement in the rotor recess (check prior to nonarming test)	Visual & Manual
2. Rotor not locked in the unarmed position (see 6.2.2) (to be checked immediately prior to assembly into fuze)	Visual & Manual
MAJOR AQL = 0.40%	
101. Length from base to movement plate exceeds max.	Gage
102. Protrusion of screws, rotor shaft and pins above plate exceeds maximum	Gage
103. Staking of plate screw(s) inadequate	Visual
104. Metal burred, damaged or defective	Visual
105. Foreign matter	Visual

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4.3.2.25 Fuze, assembly, Drawing F10542897.

Defect	Method of Inspection
CRITICAL	
1. Fuze not set on safe "S" (to be checked immediately prior to packaging)	Visual
2. Safety adapter rotor armed or partially armed as viewed through the flash hole (to be checked immediately prior to packaging)	Visual
MAJOR AQL = 0.40%	
101. Total length exceeds max.	Gage
102. Length from base to flange exceeds max.	Gage
103. Major diameter of fuze body external thread improper	Gage
104. Pitch diameter of fuze body external thread, improper	Gage
105. Major diameter of safety adapter external thread, improper	Gage
106. Pitch diameter of safety adapter external thread, improper	Gage
107. Insecure, improper or incomplete assembly	Manual
108. Threads damaged or defective	Visual
109. Dust cover (over flash hole) missing	Visual
110. Marking missing, incorrect or illegible	Visual

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4.3.2.26 Box, packing, ammunition, Drawing C8861213.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR AQL = 2.5%	
201. Contents loose	Visual
202. Marking, missing, incorrect or illegible	Visual

4.3.2.27 Box, ammunition, metal, Drawing F8864492.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Moisture inside box (check prior to packing)	Visual
102. Gasket damaged or unseated (check prior to packing)	Visual
103. Polystyrene supports/fuzes packed inadequately	Visual & Manual
104. Dessicant bags/containers damaged, loose or missing	Visual & Manual
105. Box improperly closed and secured	Visual & Manual
MINOR AQL = 2.5%	
201. Contents loose	Manual
202. Box damaged, deformed or punctured	Visual
203. Marking missing, incorrect or illegible	Visual

4.3.2.28 Process control audit-periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of the firing arm upright is not permitted at any time.

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4.3.2.28 (Cont'd)

a. Fuze subassembly, less safety adapter, Drawing F10542898.

(1) Application of lubricant
(2) Application of sealant
(3) Insertion of dessicant pellet and cushion
(4) Insertion of "O" ring between head and lower cap
(5) Head torque operation
(6) Fuze setting torque

b. Safety adapter assembly, Drawing F8596049.

(1) Application of lubricant

c. Body, movement and lower cap assembly, Drawing F10542899.

(1) Application of lubricant
(2) Insertion of "O" ring between lower cap and body
(3) Lockring torque operation
(4) Zero setting procedure
(5) Drilling operation for lockring setscrew
(6) Application of sealant

d. Movement assembly, Drawing F10523480.

(1) Locknut torqued prior to timing disc torque test

e. Movement subassembly, Drawing F10523481.

(1) Application of lubricant
(2) Top and bottom assembly screws staked prior to dynamic regulation
(3) Mainspring wound before assembly of screw
(4) Dynamic regulation operation
(5) Movement full wound after dynamic regulation

f. Laminated plates No. 1 and No. 2 assembly, Drawing F10542853.

(1) Application of lubricant

g. Escapement lever assembly Drawing F10542852 securing escapement lever spring.

4.3.3 Inspection testing. - The following tests shall be performed in accordance with the provisions of 4.4.

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4.3.3.1 Hardness (destructive). - Twenty (20) samples from each lot of the following components shall be subjected to the hardness test. Where plating is required, the sample parts tested shall be unplated, where feasible. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of components, as applicable (see 4.4.1).

Components	Drawing Number
Arbor, Escapement	D10535526
Gear, Escapement	D10542765
Pinion, Escapement	D10521101
Pinion, No. 1	D10521119
Pinion, No. 2	D10521120
Pinion, No. 3	D10521118
Pinion, No. 4	C10521123
Wheel, Escape (safety adapter assembly)	C8595539
Arm, Firing (see 4.3.3.4)	D10521116
Guide, Adjusting	C8594273
Lever, Escapement	F10520340
Spring, Hammer	C10521135
Spring, Timing Disc (see 4.3.3.2)	C10521161
Washer, Bearing	B10520292
Washer, Timing Disc	B8594267
Washer, Torque	C10520291

4.3.3.2 Load - timing disc spring (destructive). - The same twenty (20) timing disc springs subjected to the hardness test shall be subjected to the load specified on Drawing C10521161. Failure of any spring to comply with the requirements shall be cause for rejection of the lot of timing disc springs (see 4.4.2).

4.3.3.3 Compression - springs (destructive). - Twenty (20) firing pin springs from each lot shall be subjected to the compression specified on Drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of firing pin springs (see 4.4.3).

4.3.3.4 Bend - firing arms (destructive). - The same twenty (20) firing arms that comply with the hardness requirement shall be subjected to the bend specified on Drawing D10521116. Failure of any firing arm to comply with the requirement shall be cause for rejection of the lot of firing arms (see 4.4.4).

4.3.3.5 Pull test - escapement spring (nondestructive). - Twenty-five (25) escapement springs from each lot of escapement lever assemblies shall be subjected to the pull test specified on Drawing F10542852. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of escapement lever assemblies (see 4.4.5).

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4.3.3.6 Static load and torque (destructive)

4.3.3.6.1 Static load. - Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.6).

Static Load	Drawing Number
Disc and Stud Assembly	C8593103
Escapement Gear and Pinion Assembly	C10542766
Escapement Lever Assembly	F10542852
Firing Arm Assembly (test pin and arm)	C10521115
No. 1 Gear and Pinion Assembly	D10521102
No. 2 Gear and Pinion Assembly	C10521104
No. 3 Gear and Pinion Assembly	C10521106
No. 4 Gear and Pinion Assembly	C10521111
No. 5 Gear and Arbor Assembly	C10522355
Pallet Lock Assembly	B10521168
Lever Assembly (pin and spring)	B10521148
Escape Wheel Assembly (safety adapter assy)	C8595538
No. 1 Gear and Pinion Assembly (safety adapter assy)	C8594673
No. 2 Gear and Pinion Assembly (safety adapter assy)	C8594681
Balance Staff Assembly	D8594614

4.3.3.6.2 Torque. - Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot subassemblies, as applicable (see 4.4.7).

Torque	Drawing Number
Escapement Lever Assembly	F10542852
No. 5 Gear and Arbor Assembly	C10522355

4.3.3.7 Setting pin load (nondestructive). - The setting pin of fifty housing and hammer spring assemblies from each lot shall be subjected to the load specified on Drawing C10520266. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of housings (see 4.4.8).

4.3.3.8 Hammerspring deflection (nondestructive). - The hammerspring of each housing and hammerspring assembly shall be subjected to the load specified on Drawing C10520266. Failure to comply with the requirements shall be cause for rejection of the assembly (see 4.4.9).

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4.3.3.9 Kinetic energy of firing pin (nondestructive). - Fifty (50) movement subassemblies from each lot shall be tested to determine compliance with the requirement specified on Drawing F10523481. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of movement subassemblies (see 4.4.11).

4.3.3.10 Spin function - (movement subassembly) (nondestructive). - The pallet lock and safety lever assemblies within each movement subassembly shall be tested to determine compliance with the function requirement of Drawing F10523481. Failure to comply with the requirement shall be cause for rejection of the subassembly (see 4.4.12).

4.3.3.11 Spin nonfunction (movement subassembly) (nondestructive). - The movement subassembly shall be tested to determine compliance with the nonfunction requirement of Drawing F10523481. Failure of the lever lock slides to remain in the locked position shall be cause for rejection of the subassembly (see 4.4.13).

4.3.3.12 Timing disc torque (nondestructive). - The timing disc of each movement assembly shall be subjected to the torque specified on Drawing F10523480. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.14).

4.3.3.13 Arming-safety adapter assembly (nondestructive). - Each safety adapter shall be tested to determine compliance with the arming requirement of Drawing F8596049. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.15).

4.3.3.14 Nonarming-safety adapter assembly (nondestructive). - Each safety adapter shall be tested to determine compliance with the nonarming requirement of Drawing F8596049. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.16).

4.3.3.15 Environmental Tests (lots) (destructive). - Twenty-six (26) fuze assemblies from each production lot shall be subjected to the following MIL-STD tests, with the fuzes set safe "S".

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.4.1	4.4.17.1	4.2.5.1.1
Jumble	4	3.4.2	4.4.17.2	4.2.5.1.1
Forty-Foot Drop	5	3.4.3	4.4.17.3	4.2.5.1.2
Waterproofness	8	3.4.4	4.4.17.4	4.2.5.1.3
Transportation-vibration	5	3.4.5	4.4.17.5	4.2.5.1.4

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4.3.3.15.1 Acceptance criteria. - Lot acceptance criteria shall be as specified in 4.2.5.1.

4.3.3.16 Ballistics (destructive). - Sixty (60) fuzes from each lot shall be subjected to the following ballistic test phases (see 4.4.18).

Phase	Quantity	Requirement	Test Method
Impact Nonfunction Distance	20	3.5.1 & 2	4.4.18.1
Time-to-Burst			
155MM Howitzer charge 1	20	3.5.1 & 3	4.4.18.2
105MM Howitzer charge 7	20	3.5.1 & 3	4.4.18.2

4.3.3.16.1 Acceptance criteria. - The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature burst in any phase (see 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Distance Phase.

c. Any of the following in the Ambient Time-to-Burst Phase at any time setting.

(1) More than one time failure (6.3.2)

(2) More than one off times (6.3.3)

(3) Failure of the sample statistics to comply with the limits specified in Table I.

4.3.3.17 Container airtightness. - A sample quantity of packed fuze containers shall be tested for airtightness to determine compliance with the requirements of Drawing F8864492. Sampling quantities shall be determined in accordance with MIL-STD-105 using an AQL of .40 percent or MIL-STD-1235, CSP-1 using an AQL of .40 percent (see 4.4.19).

4.3.4 Inspection equipment. - The examinations and tests shall be made using equipment listed in IEL-10542897. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4. Test methods and procedures

4.4.1 Hardness. - Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three (3) hardness readings taken on each sample unit shall be used as the individual value.

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4.4.2 Load (timing disc spring). - The test shall be performed using the values as specified on Drawing C10521161. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.3 Compression. - The firing pin springs shall be compressed for the length of time specified on Drawing C10521132. The springs that have been subjected to the compression test shall be scrapped.

4.4.4 Bend. - The firing arm bend test shall be performed as specified on Drawing D10521116. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.5 Pull test (escapement spring). - The test shall be performed using the values as specified on Drawing F10542852.

4.4.6 Static Load. - Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicalbe drawing. All samples tested shall be scrapped.

4.4.7 Torque. - Escapement lever assemblies shall be tested prior to insertion of the escapement spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

4.4.8 Setting pin load. - The setting pin within the housing and hammerspring assemblies shall be tested for the minimum load requirement of Drawing C10520266. The dimension from the bottom of the pin to surface "C" shall be checked before and after load application.

4.4.9 Hammerspring deflection (housing and hammerspring assembly). - The required load shall be applied at the specified locations parallel to the vertical axis of the assembly as specified on Drawing C10520266. The hammerspring must return to surface "C" when released.

4.4.10 Dynamic regulation monitoring. - The following procedure shall be observed: The movement subassemblies shall be regulated, using the operational procedures furnished with the equipment. The movement subassemblies shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6,000 ± 15 rpm. Timekeeping and performance shall be checked over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run. Inspection shall be performed on a monitoring basis only (paragraph 4.3.2.28).

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4.4.11 Kinetic energy of firing pin (movement subassembly). - The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position.

4.4.12 Spin function (movement subassembly). - The function test shall be performed on completely wound subassemblies. The subassemblies shall be rotated at the rpm specified on Drawing F10523481 in the clockwise direction when viewed from the bottom, for a period of not less than 10 seconds at constant rpm to determine compliance with the requirement. Compliance with the requirements shall be determined after the test by the position of the safety lever (contacting the periphery of the collar). At the conclusion of the test, each mechanism shall be rewound. The function test shall be performed prior to the nonfunction test.

4.4.13 Spin nonfunction (movement subassembly). - The nonfunction test shall be performed on completely wound subassemblies. The subassemblies shall be rotated at the rpm specified on Drawing F10523481 in a clockwise direction when viewed from the bottom, for a period of not less than three (3) seconds at constant rpm to determine compliance with the requirement. (Neither lever lock slide releases the escapement lever.) This test shall be the last spin test to which the movement is subjected.

4.4.14 Timing disc torque. - The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance with the requirements of Drawing F10523480.

4.4.15 Arming (safety adapter assembly). - The unarmed safety adapter shall be rotated in a counterclockwise direction when viewed from the movement plate end, at the rpm specified on Drawing F8596049. The arming test shall be performed prior to the nonarming test. At the conclusion of the test, the rotor shall be returned to the unarmed position (see 6.2.1 and 6.2.2).

4.4.16 Nonarming (safety adapter assembly). - The unarmed safety adapter (see 6.2.2) shall be rotated in a counterclockwise direction when viewed from the movement plate end, at the rpm specified on Drawing F8596049. The nonarming test shall be the last spin test to which the safety adapter is subjected prior to assembly into the fuze.

4.4.17 Environmental test procedures. - The component parts of the fuze, which have been disassembled and inspected following tests, shall be scrapped. Where spin testing is required, the nonarming test shall be performed prior to the arming test (Drawing F10542897).

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4.4.17.1 Jolt. - The test shall be performed in accordance with the procedure specified in MIL-STD-350. The fuze shall be disassembled and inspected as follows:

a. Remove dust seal and check position of adapter rotor.

b. Remove head assembly and check movement safety disc relative to the firing arm.

c. Remove safety adapter. Check detents and position of rotor.

d. Remove closing plug, check relay.

e. Remove lower cap and housing. Check timing disc retainer.

f. Remove movement assembly, from body. Check position of firing pin, pallet lock slides, check M47 detonator.

g. Disassemble subassemblies to the extent that all critical parts can be examined; i.e., all gears, pinions, and springs, etc., for damage.

4.4.17.2 Jumble. - The test shall be performed in accordance with the procedure specified in MIL-STD-351. The fuze shall be disassembled and inspected as outlined in paragraph 4.4.17.1.

4.4.17.3 Forty-foot drop. - The test shall be performed in accordance with the procedure specified in MIL-STD-352. The fuze shall be disassembled and inspected as follows:

a. Remove dust seal and check position of adapter rotor.

b. Remove safety adapter and move rotor to armed position, check detonator.

4.4.17.4 Waterproofness. - The test shall be performed in accordance with the procedure specified in MIL-STD-314. The safety adapter flash hole shall be adequately sealed for the test. The fuze shall be disassembled and inspected as follows:

a. Remove safety adapter, head, lower cap, check all visible internal surfaces for evidence of water entry.

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4.4.17.5 Transportation-vibration. - The test shall be performed in accordance with the procedure specified in MIL-STD-353 as follows:

a. Parallel to longitudinal axis, nose up.

b. First transverse axis to be parallel to a line running through the 100 second markings.

c. Second transverse axis to be parallel to a line running through the 70 second markings. The fuzes subjected to this test shall be disassembled and inspected as follows:

(1) Remove dust seal and check position of adapter rotor.

(2) Remove head assembly and check movement safety disc relative to the firing arm.

(3) Remove safety adapter. Spin for nonarming and arming requirements.

(4) Remove closing plug, check relay.

(5) Remove lower cap and housing. Check timing disc retainer.

(6) Remove movement assembly from the body. Check position of firing pin and M47 detonator. Spin for nonfunction and function requirements.

(7) Disassemble subassemblies to the extent that all critical parts can be examined, i.e., all gears, pinions, and springs, etc., for damage.

4.4.18 Ballistics. - The complete fuze assembly, Drawing F10542897 shall be assembled to a suitable projectile and fired as follows (see 6.3.4).

4.4.18.1 Impact nonfunction distance phase. - The sample fuzes shall be fired as specified in 3.5.2 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.16 as appropriate. Observation of impact element function is required.

4.4.18.2 Time-to-Burst Phase. - The sample fuzes shall be fired as specified in 3.5.3 under the conditions delineated in 4.2.6 or 4.3.3.16 as appropriate. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The average of the two values shall be rounded off to the second decimal place using the method prescribed by the American Society for Testing Materials.

4.4.18.2.1 The sample mean and standard deviation values shall be calculated by:

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$$\bar{X} = \frac{\Sigma X}{n}$$

$$\sigma = \sqrt{\frac{n\Sigma X^2 - (\Sigma X)^2}{n\ (n-1)}}$$ or equivalent

Where:

X is the individual value

n is the number of values

All values of the sample shall be used to calculate the sample mean and standard deviations. The standard deviation value obtained shall then be used in determination of outliers (non-normal values) by using the criteria $\bar{X}$ plus and minus four (4) standard deviations. A value judged to be an outlier shall be excluded and the remaining values in the sample used to recalculate $\bar{X}$ and σ.

4.4.18.3 Excess pressure phase. - The sample fuzes shall be fired as specified in 3.5.4 under the conditions delineated in 4.2.6. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.18.4 Extreme temperature phase. - The sample fuzes shall be fired as specified in 3.5.3 under the conditions delineated in 4.2.6. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.18.4.1 Temperature conditioning (-65° F and +145° F). - The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus 65° F or plus 145° F as applicable, until stabilization of the fuze temperature is reached (see 6.4). After stabilization, the fuze shall be saturated at minus 65° F or plus 145° F as applicable, for one hour, minimum.

4.4.19 Container airtightness

The packed fuze container shall be subjected to an external air pressure of three (3) pounds per square inch, gage, for a period of not less than 15 seconds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level "A".

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5.1.1 Level "A" - Packaging shall be in accordance with Drawing F8864492.

5.2 Packing

5.2.1 Level "A"- Unit boxes shall be packed in accordance with Drawing C8861213.

5.3 Marking. - Ammunition boxes shall be marked in accordance with Drawing F8864492. Boxes shall be marked in accordance with Drawing C8861213. The ICC marking shall be "PERCUSSION FUZES - HANDLE CAREFULLY".

6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 or those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/unarmed safety adapter

6.2.1 Armed. - Rotor is locked by the spring loaded lock pin in the position in which the detonator is fully visible through the flash hole.

6.2.2 Unarmed. - Rotor to remain locked by both spring loaded detents with the detonator not all visible through the flash hole.

6.3 Ballistic parameters

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6.3.1 Premature burst.- Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun.

b. The round detonates before 45 seconds when set at 60 seconds or before 1.9 second when set at 3 seconds.

c. The round detonates before impact when set on safe "S" in the Impact Nonfunction Distance Phase.

6.3.2 Time failure.- A time failure is considered to have occurred if the round fails to detonate during flight.

6.3.3 Off-time.- An off-time is an individual round burst time not within the limits for individual rounds specified in Table I.

6.3.4 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated fuze may be used for this purpose.

6.5 Data Cards.- Data card information shall be as specified in MIL-STD-1167.

6.6 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

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